

Circuit Simulation Lab:31

Parity Generator and Checker

The purpose of this experiment is to design a Parity Generator and Checker. A Parity Generator is a combinational logic circuit that generates the parity bit in the transmitter. On the other hand, a circuit that checks the parity in the receiver is called Parity Checker. A combined circuit or device of parity generators and parity checkers are commonly used in digital systems to detect the single bit errors in the transmitted data.

```
module parity_generator(  
input [7:0] data,  
  
output even_parity,  
output odd_parity  
);  
  
assign even_parity = ^data;  
assign odd_parity = ~(^data);
```

```
Endmodule
```

```
TESTBENCH  
module tb_parity_generator;  
  
reg [7:0] data;  
  
wire even_parity;  
wire odd_parity;  
  
parity_generator dut(  
.data(data),  
.even_parity(even_parity),  
.odd_parity(odd_parity)  
);  
  
initial begin  
  
$dumpfile("parity_gen.vcd");  
$dumpvars(0,tb_parity_generator);  
  
data = 8'b00000000; #10;
```

```

data = 8'b00000001; #10;
data = 8'b10101010; #10;
data = 8'b11111111; #10;

$finish;

end

initial
$monitor(
"time=%0t data=%b even=%b odd=%b",
$time,data,even_parity,odd_parity
);

Endmodule

```

